

NIELIT CCC • UNIT 04

Spreadsheet (LibreOffice Calc)

Curriculum Hours: 12h (4h Theory + 8h Lab) | Topics: 9 | Language: English

UNIT OVERVIEW & PEDAGOGICAL SCOPE

A spreadsheet is an interactive digital application for organizing, calculating, analyzing, and visually presenting numerical data arranged in a two-dimensional grid of rows and columns. LibreOffice Calc is the standardized FOSS spreadsheet component taught in the NIELIT CCC curriculum. This unit covers the Calc grid matrix, cell addressing rules, data manipulation, mathematical formulas, built-in statistical functions, cell referencing modes, AutoFill series, sorting and filtering, chart creation, diagnostic error codes, and master keyboard shortcuts.

WHAT YOU WILL MASTER IN THIS UNIT

- The architectural dimensions of LibreOffice Calc (1,048,576 Rows, 1,024 Columns A to AMJ, 1,073,741,824 Cells, .ods extension, Zoom range 20% to 400%).
- Cell addressing rules and automatic data alignment (Text = Left, Numbers/Dates = Right, Booleans = Center, Single quote ' prefix).
- Cell and row manipulation: In-cell editing (F2), Format Cells (Ctrl+1), In-cell line break (Ctrl+Enter), and Insert/Delete rows (Ctrl + + / Ctrl + -).
- Formulas and functions: Mandatory '=' prefix, operator precedence, and essential functions (SUM, AVERAGE, COUNT vs COUNTA, MAX, MIN, NOW vs TODAY).
- Cell referencing modes: Relative (A1), Absolute (\$A\$1), Mixed (\$A1 / A\$1), and toggling references (Shift+F4).
- Data organization: AutoFill drag handle, AutoFilter (Ctrl+Shift+L), multi-column sorting, and Freeze Panes.
- The 10 standard chart types (Column, Bar, Pie, Line...) and the 4-step Chart Wizard.
- Diagnostic formula error codes: ###, #DIV/0!, #NAME?, #VALUE!, #REF!.
- Complete Master Shortcut Table for LibreOffice Calc.

TOPIC 4.1: INTRODUCTION TO SPREADSHEETS & CALC INTERFACE

LibreOffice Calc is an advanced OpenDocument spreadsheet processor designed to handle numerical data, financial models, mathematical formulas, statistical analysis, and interactive business charts.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Spreadsheets automate numerical calculations, eliminate human arithmetic errors in accounting and budgeting, and dynamically recompute complex formulas whenever input values change.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

A Calc document is called a **Workbook**, which contains one or more individual **Worksheets**. Each sheet is an expansive grid where horizontal **Rows** and vertical **Columns** intersect to form distinct rectangular **Cells**.

CALC GRID DIMENSIONS & SYSTEM DEFAULTS

- Total Number of Rows: Exactly 1,048,576 rows (numbered sequentially from 1 to 1048576).
- Total Number of Columns: Exactly 1,024 columns (lettered from A to Z, then AA to ZZ, ending at column AMJ).
- Total Number of Cells: 1,073,741,824 individual cells per worksheet.
- Default File Extension: `.ods` (OpenDocument Spreadsheet).

- Default Document Name: 'Untitled 1' (until saved).
- Zoom Percentage Range: Minimum zoom is 20%; Maximum zoom is 400%.
- Name Box: Located on the left above the sheet grid; displays the address of the currently active cell (e.g. 'A1').
- Formula Bar (Input Line): Located to the right of the Name Box; displays the actual underlying formula or text of the selected cell.

Launch Calc. Look at the top left Name Box: It shows 'A1'. The last column on the far right is labeled 'AMJ'. The last row at the very bottom is '1048576'. Click into cell 'C5'; the Name Box immediately updates to show 'C5'.

- Total number of Rows in LibreOffice Calc: 1,048,576 (2^{20}).
- Total number of Columns in LibreOffice Calc: 1,024 (2^{10}).
- The last column letter in LibreOffice Calc is AMJ.
- Default file extension for LibreOffice Calc is '.ods'.
- The zoom percentage range in LibreOffice Calc is 20% to 400%.

QUICK TOPIC RECAP

Calc matrix: 1,048,576 Rows, 1,024 Columns (A to AMJ), .ods extension, Zoom 20%-400%. Name Box shows active cell address; Formula Bar displays cell contents.

TOPIC 4.2: CELL ADDRESSING AND ENTERING DATA

A cell address (cell reference) is the unique coordinate identifier formed by combining the column letter and row number (e.g., column 'B' and row '10' = 'B10').

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Entering correct data types and understanding default cell alignments is essential for ensuring mathematical formulas calculate numerical values properly.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

When you type into a cell, Calc automatically detects the data type and applies default horizontal alignments. If a number is prefixed with a single quote ('), Calc forces it to be treated as plain text.

CELL EDITING & FORMATTING SHORTCUTS

- In-Cell Edit Mode: Select a cell and press 'F2' to place the blinking text cursor directly inside the cell to modify text without overwriting existing content.
- Format Cells Dialogue Box: Press 'Ctrl + 1' to open the full cell formatting dialogue (Numbers, Decimals, Currency, Borders, Font, Alignment).
- Current Date Shortcut: Press 'Ctrl + ;' (Semicolon) to insert today's static date into the active cell.
- Current Time Shortcut: Press 'Ctrl + Shift + ;' to insert the current static time.

DATA TYPES AND DEFAULT HORIZONTAL ALIGNMENTS IN CALC

Data Type	Examples Entered	Default Horizontal Alignment	Behavior in Formulas
Text / Labels	Name, Product, Jaipur	Left Aligned	Ignored by numerical calculation functions like SUM()
Numbers / Values	450, -89.5, 98.75	Right Aligned	Treated as numeric values for mathematical operations
Dates & Times	25/08/2026, 14:30:00	Right Aligned	Stored internally as serial numbers representing elapsed days
Boolean Logic	TRUE, FALSE	Center Aligned	Represents logical 1 (TRUE) and 0 (FALSE) conditions

Text-Forced Numbers	'12345, '0098	Left Aligned	Single quote (') forces number to behave as non-numeric text
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If you type a mobile number `0987654321` into a numeric cell, Calc automatically drops the leading zero and aligns it right as `987654321`. If you type ``0987654321` (with a single quote prefix), Calc keeps the leading zero and aligns it left as text.

- In LibreOffice Calc: Text is Left-aligned; Numbers/Dates are Right-aligned; Booleans are Center-aligned.
- Shortcut key to edit the active cell: `F2`.
- Shortcut key to open the Format Cells dialogue box: `Ctrl + 1`.
- A single quote (``) prefix forces numeric digits to be stored as text.
- Shortcut for Current Date: `Ctrl + ;` | Shortcut for Current Time: `Ctrl + Shift + ;`.

QUICK TOPIC RECAP

Cell address = Column+Row (e.g. B5). Alignments: Text = Left, Numbers = Right, Booleans = Center. Edit cell = F2, Format Cells = Ctrl+1, Date = Ctrl+;, Time = Ctrl+Shift+;.

TOPIC 4.3: MANIPULATING CELLS, ROWS, AND COLUMNS

Cell, row, and column manipulation includes inserting, deleting, resizing, hiding, merging, and wrapping content within the spreadsheet grid.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Table layouts frequently change as new data is collected. Manipulating grid geometry allows spreadsheets to adjust dynamically without corrupting existing formulas.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Calc allows users to insert new rows/columns or delete unwanted rows. When rows are inserted, all existing formulas automatically adjust their cell coordinates.

ROW AND COLUMN MANIPULATION SHORTCUTS

- Insert Rows / Columns: Press `Ctrl + +` (Control with Plus key on numeric keypad), or right-click column/row header -> Insert.
- Delete Rows / Columns: Press `Ctrl + -` (Control with Minus key on numeric keypad), or right-click -> Delete.
- In-Cell Line Break: Press `Ctrl + Enter` while typing inside a cell to start a new line within the same single cell without moving to the cell below.
- Select Entire Row: Press `Shift + Spacebar`.
- Select Entire Column: Press `Ctrl + Spacebar`.
- Select Entire Worksheet: Press `Ctrl + A` (or `Ctrl + Shift + Spacebar`).
- Wrap Text: Enables automatic text wrapping so long sentences expand the row height rather than spilling into adjacent empty cells.

You are entering a multi-line address in cell A1: Type 'House No. 42', press `Ctrl + Enter` (in-cell line break), type 'M.G. Road', press `Ctrl + Enter`, and type 'New Delhi'. All three lines reside cleanly inside the single cell A1.

- Shortcut key to Insert Cells/Rows/Columns: `Ctrl + +`.
- Shortcut key to Delete Cells/Rows/Columns: `Ctrl + -`.
- Shortcut key for in-cell line break (multi-line text in single cell): `Ctrl + Enter`.
- Select Entire Row: `Shift + Spacebar` | Select Entire Column: `Ctrl + Spacebar`.

QUICK TOPIC RECAP

Insert = Ctrl++, Delete = Ctrl+-. Multi-line in cell = Ctrl+Enter. Select row = Shift+Space, select column = Ctrl+Space. Formulas automatically shift on row insertion.

TOPIC 4.4: FORMULAS, FUNCTIONS, AND CALCULATIONS

A **Formula** is a user-defined mathematical expression that performs calculations on cell values (e.g. `=A1+B1`). A **Function** is a built-in, pre-programmed calculation module (e.g. `=SUM(A1:A10)`).

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Formulas and functions form the core engine of spreadsheets, executing complex financial, statistical, and logical calculations instantaneously across thousands of records.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

All formulas and functions in LibreOffice Calc **must** begin with an equal sign (`=`). Without the `=` prefix, Calc treats the input as plain text.

MATHEMATICAL OPERATORS & EVALUATION PRECEDENCE

- Arithmetic Operators: Exponentiation (`^`), Multiplication (`*`), Division (`/`), Addition (`+`), Subtraction (`-`).
- Order of Precedence (BODMAS): Parentheses `()` are evaluated first, followed by Exponents `^`, then Multiplication/Division `*` `/`, and finally Addition/Subtraction `+` `-`.
- Comparison Operators: Equal (`=`), Not Equal (`<>`), Greater Than (`>`), Less Than (`<`), Greater Than or Equal (`>=`), Less Than or Equal (`<=`).

ESSENTIAL BUILT-IN FUNCTIONS IN LIBREOFFICE CALC

Function Name	Syntax & Example	Calculated Mathematical Output	Key Exam Distinction
SUM()	<code>=SUM(A1:A5)</code>	Adds all numbers in the specified cell range	Ignores text cells within the specified range
AVERAGE()	<code>=AVERAGE(A1:A5)</code>	Calculates the arithmetic mean of numeric values	Divides sum of numbers by count of numeric cells
COUNT()	<code>=COUNT(A1:A5)</code>	Counts only cells that contain numbers	Strictly ignores blank cells and text strings
COUNTA()	<code>=COUNTA(A1:A5)</code>	Counts all non-empty cells (numbers + text)	Counts any cell that is not completely blank
MAX() / MIN()	<code>=MAX(A1:A5) / =MIN(A1:A5)</code>	Returns the highest / lowest numerical value in range	Quickly identifies peak or minimum marks/sales
NOW()	<code>=NOW()</code>	Returns current system date and current time	Dynamic: Updates every time the sheet recalculates
TODAY()	<code>=TODAY()</code>	Returns current system date only (without time)	Dynamic: Updates daily based on system clock
ROUND()	<code>=ROUND(45.678, 2)</code>	Rounds a number to specified decimal places (45.68)	Uses standard mathematical rounding rules

Evaluating the formula `=10 + 5 * 2`: Calc calculates multiplication first (`5 * 2 = 10`), then adds 10, producing `=20`. If written as `=(10 + 5) * 2`, the parentheses evaluate first (`15 * 2`), producing `=30`.

- Every formula in LibreOffice Calc MUST begin with an equal sign (=).
- COUNT() counts only numeric cells; COUNTA() counts all non-empty cells.
- NOW() returns current date and time; TODAY() returns current date only.
- The 'Not Equal To' operator in Calc is <>.

QUICK TOPIC RECAP

Every formula starts with =. BODMAS rules apply. SUM adds numbers, AVERAGE finds mean, COUNT counts numbers, COUNTA counts all non-blank cells. NOW() gives date+time.

TOPIC 4.5: CELL REFERENCING: RELATIVE, ABSOLUTE, AND MIXED

Cell referencing defines how cell addresses behave when a formula is copied or dragged using the AutoFill handle from one cell to other adjacent cells.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Understanding cell referencing is necessary to prevent calculation errors when copying formulas down a column containing fixed tax rates or discount percentages.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Prepend a dollar sign (\$) before a column letter or row number to lock that coordinate. The Shift + F4 shortcut cycles through reference modes.

TOGGLING CELL REFERENCES VIA SHIFT + F4

- When typing or editing a formula: Select or place the cursor on a cell reference (e.g., A1) and press Shift + F4.
- Cycle Sequence: A1 (Relative) -> \$A\$1 (Absolute) -> A\$1 (Row Fixed) -> \$A1 (Column Fixed) -> A1.

THE THREE CELL REFERENCING MODES

Referencing Mode	Syntax Example	Behavior When Formula Is Copied	Practical Application Scenario
Relative Reference	A1	Both column and row change relative to the new position (e.g. copying down makes A2, A3)	Calculating individual line-item totals (Price * Qty) down a column
Absolute Reference	\$A\$1	Both column and row remain completely locked and fixed regardless of where copied	Multiplying every row by a single fixed GST Tax Rate cell in \$E\$1
Mixed Reference (Row)	A\$1	Column changes when copied horizontally, but Row 1 remains locked vertically	Multiplication tables where row headers remain fixed
Mixed Reference (Col)	\$A1	Column A remains locked, but row number changes when copied vertically	Multi-column tables referencing a fixed baseline column A

In cell C2, you write =A2 * \$E\$1, where A2 is the product price and E1 is the fixed 18% GST rate. When you drag this formula down to C3 and C4, it automatically becomes =A3 * \$E\$1 and =A4 * \$E\$1. The product row increments, but the tax cell remains locked to E1.

- A dollar sign (\$) indicates an Absolute (locked) cell reference (e.g. \$A\$1).
- Shortcut key to cycle between Relative, Absolute, and Mixed referencing: Shift + F4 (in Calc).
- Relative referencing (e.g. A1) is the default mode in LibreOffice Calc.

QUICK TOPIC RECAP

Relative A1 shifts when copied. Absolute \$A\$1 stays locked. Mixed \$A1 or A\$1 locks one coordinate. Press Shift + F4 to cycle referencing modes.

TOPIC 4.6: AUTOFILL, SORTING, AND FILTERING DATA

AutoFill automatically populates sequential data series, while Sorting reorders rows and Filtering temporarily isolates specific records matching designated criteria.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

AutoFill saves hours of repetitive typing, while sorting and filtering allow users to extract meaningful insights from large tables containing thousands of entries.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

AutoFill uses the small black square handle in the bottom-right corner of the active cell. AutoFilter attaches interactive dropdown arrows to table headers.

DATA MANAGEMENT TOOLS IN CALC

- AutoFill Drag Handle: Position the cursor on the tiny square at the bottom-right corner of the selection (cursor turns to a '+' crosshair) and drag across rows/columns.
- Automatic Built-in Series: Numbers (type 1, 2, highlight both and drag), Weekdays (Monday, Tuesday...), Months (January, February...), Dates (01/01/2026, 02/01/2026...).
- Sorting Data: Access via 'Data -> Sort'. Sorts alphabetically (A-Z, Z-A) or numerically (Smallest to Largest, Largest to Smallest) across up to 3 criteria columns.
- AutoFilter: Press 'Ctrl + Shift + L' (or menu 'Data -> AutoFilter') to insert dropdown filter arrows on header cells.
- Freeze Panes: Menu 'View -> Freeze Rows and Columns' locks header rows at the top so they remain visible while scrolling down large sheets.

To create serial numbers 1 to 100: Type '1' in cell A1 and '2' in cell A2. Highlight both cells and double-click the AutoFill handle. Calc instantly generates numbers 1 through 100 down the column in less than a second.

- Shortcut key to toggle AutoFilter on/off: 'Ctrl + Shift + L'.
- AutoFill automatically extends days of the week, months, dates, and numeric progressions.
- Freeze Panes (View -> Freeze Rows and Columns) keeps header rows visible during scrolling.

QUICK TOPIC RECAP

AutoFill extends series via the bottom-right fill handle. AutoFilter = Ctrl+Shift+L. Sorting orders data A-Z or Z-A. Freeze Panes locks headers.

TOPIC 4.7: CHARTS AND VISUAL DATA REPRESENTATION

Charts are graphical representations of worksheet data that transform complex numerical tables into visual diagrams such as bars, lines, and pie slices.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Visual charts make trends, proportions, comparisons, and financial forecasts immediately understandable to business managers and students.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Select the data table, click the Chart icon or select 'Insert -> Chart', and complete the 4-step Chart Wizard to generate an embedded vector graphic chart.

THE 4-STEP CHART WIZARD PROCESS (INSERT -> CHART)

- Step 1: Chart Type (Choose from 10 chart categories: Column, Bar, Pie, Line, etc.).

- Step 2: Data Range (Verify selected table range and indicate whether data series is in rows or columns).
- Step 3: Data Series (Customize individual legend labels and categories).
- Step 4: Chart Elements (Add Chart Title, Subtitle, X-Axis label, Y-Axis label, and Legend position).

THE 10 STANDARD CHART TYPES IN LIBREOFFICE CALC

Chart Type	Visual Appearance	Best Analytical Use Case
Column Chart	Vertical rectangular bars	Comparing discrete categories across different groups (e.g. quarterly sales by region)
Bar Chart	Horizontal rectangular bars	Comparing categories with long label names or displaying ranked items
Pie Chart	Circular disc divided into percentage slices	Showing proportional parts of a single whole (100% total distribution)
Line Chart	Data points connected by continuous lines	Displaying continuous trends over time (e.g. monthly temperature, stock trends)
Area Chart	Shaded areas beneath line curves	Emphasizing the magnitude of volume change over time
Scatter (X-Y)	Unconnected or connected point coordinates	Scientific engineering data showing statistical correlation between 2 variables
Bubble Chart	Varying sized circular bubbles	Comparing three data variables simultaneously (X, Y, and Bubble Size)
Net (Radar)	Spider-web grid extending from center	Comparing performance profiles across multiple qualitative criteria
Stock Chart	High-Low-Close candle bars	Visualizing financial stock market price fluctuations over trading days
Column & Line	Combination of vertical bars and line graphs	Displaying two related metrics with different scales (e.g. Revenue & Profit %)

To display an annual family budget: Select the expense categories and amounts, launch the Chart Wizard, and choose 'Pie Chart'. Calc automatically draws a colored circle showing food, rent, and electricity as percentage slices adding up to 100%.

- A Pie Chart is best suited for showing parts of a single whole (percentage proportions).
- A Line Chart is best suited for tracking continuous trends over time.
- The Chart Wizard in LibreOffice Calc consists of exactly 4 steps.
- LibreOffice Calc supports 10 distinct standard chart types.

QUICK TOPIC RECAP

Calc offers 10 chart types via a 4-step wizard. Pie chart = percentage parts of a whole. Line chart = trends over time. Column/Bar = category comparisons.

TOPIC 4.8: COMMON FORMULA ERRORS AND PRINTING SPREADSHEETS

Diagnostic formula error codes are error flags displayed in a cell when a formula contains invalid math, misspelled function names, or incorrect data references.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Recognizing error codes enables users to troubleshoot broken formulas and fix accounting calculations quickly.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

When Calc encounters an invalid computational state during formula evaluation, it halts computation and writes a standardized

diagnostic error string into the cell.

SPREADSHEET PRINTING ESSENTIALS

- Page Break Preview: Menu `View -> Page Break Preview` displays where pages will split during physical printing with adjustable blue boundary lines.
- Define Print Area: Highlight specific cells -> `Format -> Print Ranges -> Define` to print only the designated data table, ignoring outer scratch calculations.
- Fit to Page: Configure in `Format -> Page Style -> Sheet` to scale large tables onto a single sheet of paper.

COMMON CALC FORMULA ERROR CODES REFERENCE

Error Code	Underlying Cause / Error Meaning	Corrective Solution
###	Column width is too narrow to display the number or date	Double-click the column header border to auto-widen the column
#DIV/0!	Attempting to divide a number by zero (or an empty cell)	Check divisor cell to ensure it is not zero or empty
#NAME?	Unrecognized function name or invalid text without quotes	Correct misspelled function (e.g. fix `=SUMM(A1:A5)` to `=SUM(A1:A5)`)
#VALUE!	Incorrect argument or operand data type (e.g. adding text to number)	Ensure cells referenced in arithmetic contain valid numbers, not text
#REF!	Invalid cell reference (e.g. a referenced row/column was deleted)	Update formula to reference an existing valid cell
#N/A	Data is not available (common in lookup formulas)	Verify lookup search term exists in the target lookup table

You enter `=50/0` in cell A1. Calc immediately displays `#DIV/0!`. When you enter `10000000` into a narrow column, Calc displays `###`. Double-clicking the line between column headers widens the column, instantly revealing `10,000,000`.

- `###` indicates the column width is too narrow to display the number.
- `#DIV/0!` indicates an illegal division by zero.
- `#NAME?` indicates a misspelled function name or unrecognized text.
- `#VALUE!` indicates an incompatible data type (e.g. mathematical addition with a text string).

QUICK TOPIC RECAP

Error codes: `###` = narrow column, `#DIV/0!` = division by zero, `#NAME?` = misspelled function, `#VALUE!` = wrong data type, `#REF!` = deleted cell reference.

TOPIC 4.9: SHORTCUT KEYS MASTER REFERENCE FOR LIBREOFFICE CALC

A master reference table of all essential keyboard shortcuts in LibreOffice Calc, categorized for rapid review before CCC examinations.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Calc shortcuts are heavily tested in NIELIT examinations and significantly increase spreadsheet calculation speed.

COMPLETE LIBREOFFICE CALC SHORTCUTS TABLE

Shortcut Key	Command Name	Action Performed inside LibreOffice Calc
F2	Edit Cell	Places text cursor inside active cell to edit contents directly
Ctrl + 1	Format Cells	Opens the complete Format Cells dialogue window
Ctrl + Enter	In-Cell Line Break	Inserts a new line within the same single cell

Ctrl + ;	Insert Current Date	Inserts static today's date into the active cell
Ctrl + Shift + ;	Insert Current Time	Inserts static current time into the active cell
Ctrl + +	Insert Cells/Rows	Opens dialogue to insert new rows, columns, or cells
Ctrl + -	Delete Cells/Rows	Deletes selected cells, rows, or columns
Shift + Spacebar	Select Entire Row	Selects the entire horizontal row containing the cursor
Ctrl + Spacebar	Select Entire Column	Selects the entire vertical column containing the cursor
Ctrl + Shift + L	AutoFilter Toggle	Applies or removes interactive filter dropdown arrows
Shift + F4	Toggle Reference	Cycles reference mode: A1 -> \$A\$1 -> A\$1 -> \$A1
Ctrl + F2	Function Wizard	Opens dialogue with all categorized built-in functions
Ctrl + Home	Go to Cell A1	Moves active cell cursor directly to the top-left cell A1
Ctrl + End	Go to Last Used Cell	Moves cursor to the bottom-rightmost cell containing data
Ctrl + F5	Toggle Sidebar	Opens or closes the collapsible formatting properties sidebar
Ctrl + Shift + O	Print Preview	Opens full-screen print preview of the worksheet
Ctrl + P	Print	Opens standard print dialogue box
Ctrl + S	Save Spreadsheet	Saves modifications to the active .ods workbook

To format a sales column quickly: Click column header, press `Ctrl + 1` to format as currency with 2 decimal places, press `Ctrl + Shift + L` to enable AutoFilter, and press `Ctrl + Home` to return instantly to cell A1.

- Edit Cell: `F2` | Format Cells: `Ctrl + 1`.
- Insert Date: `Ctrl + ;` | Insert Time: `Ctrl + Shift + ;`.
- Insert Row/Col: `Ctrl + +` | Delete Row/Col: `Ctrl + -`.
- AutoFilter: `Ctrl + Shift + L` | Toggle Reference: `Shift + F4`.
- Jump to Cell A1: `Ctrl + Home`.

QUICK TOPIC RECAP

Memorize key Calc hotkeys: F2 = Edit cell, Ctrl+1 = Format cells, Ctrl+; = Date, Ctrl+Shift+; = Time, Ctrl++ = Insert, Ctrl+- = Delete, Shift+F4 = Reference toggle, Ctrl+Shift+L = AutoFilter.

TOPIC undefined:

- LibreOffice Calc grid dimensions: 1,048,576 Rows, 1,024 Columns (A to AMJ), .ods extension, Zoom range 20% to 400%.
- Cell Addressing: Column Letter + Row Number (e.g. B5). Name Box displays active address; Formula Bar displays cell content.
- Default Alignments: Text is Left-aligned, Numbers/Dates are Right-aligned, Booleans are Center-aligned. Single quote (') forces text mode.
- Key Cell Operations: Edit cell = `F2`, Format cells = `Ctrl + 1`, In-cell line break = `Ctrl + Enter`.
- Row & Column Shortcuts: Insert = `Ctrl + +`, Delete = `Ctrl + -`, Select Row = `Shift + Spacebar`, Select Column = `Ctrl + Spacebar`.
- Every formula MUST begin with an equal sign (=). Evaluation follows standard BODMAS mathematical precedence.
- Core Functions: `SUM()`, `AVERAGE()`, `COUNT()` (numbers only), `COUNTA()` (all non-empty cells), `MAX()`, `MIN()`, `NOW()` (date+time), `TODAY()` (date only).
- Cell Referencing: Relative `A1`, Absolute `\$A\$1`, Mixed `\$A1` or `A\$1`. Toggle reference modes with `Shift + F4`.
- Data Organization: AutoFill drag handle extends series. AutoFilter toggle = `Ctrl + Shift + L`. Freeze Panes keeps header rows locked.

- Calc supports 10 chart types created in 4 steps via the Chart Wizard. Pie charts show proportions; Line charts show continuous trends.
- Diagnostic Error Codes: `###` (column narrow), `#DIV/0!` (division by zero), `#NAME?` (misspelled function), `#VALUE!` (invalid data type), `#REF!` (deleted cell reference).

KEY TERMS, ACRONYMS & OFFICIAL FULL FORMS

--- END OF UNIT 04 DETAILED STUDY BOOK • NIELIT CCC (80 HOURS) ---